

B4Slider Cheat Sheet

set card · AXIS 1–5 packed · silk 1 2 3 typical



Recommended B4Slider 3-axis panel layout

Panel

Silk: S SET · < MOVE_L · > MOVE_R · * OPTION · 1 2 3 AXIS (recommended plate). Firmware also wires AXIS_4/5. Legal chord: each key ≤ packed getAxisCount(). Invalid = blue blink. Boot = axis 1. OLED Ax 1+5. Unlock: OPTION. Homing: motors only. Shutter: SliderMC CT, not a UIC pin.

AXIS select

ACTION	KEY	DESCRIPTION
Hold AXIS keys	1...5	Selection = currently pressed; white flash N times
Release all AXIS	–	Keep last valid mask
Illegal chord	e.g. 5 on 2-axis MC	Blue blink; mask unchanged

MOVE / SET+MOVE act on the selection. Unselected MT slots are skip _ (time-synced). No AXIS_6 key.

Move

ACTION	KEY	DESCRIPTION
(MOVE_L / MOVE_R) tap ≤ 1/5s	< / > tap	Locked cruise of selection toward that soft end
(MOVE_L / MOVE_R) hold > 1/5s	< / > hold	Hold-to-run; soft-stop on release
Same MOVE tip while locked	< or > tip	Soft-stop cruise
Opposite MOVE	< or >	Reverse / retarget other soft end
OPTION + (MOVE_L / MOVE_R)	* + < / >	Same as MOVE at panel max speed
OPTION hold while moving	* hold	Boost to max speed; release → SPEED
SET while moving	S	Soft-stop
MOVE_L + MOVE_R	< >	Halt; driver off until any key
MOVE_L + MOVE_R hold ≥ 1s	< > hold ≥ 1s	Swap left/right
All four (L+R+OPTION+SET)	< > * S	Halt + reset like power-up

Tap vs hold: B4S_MOVE_TAP_MS (333 ms). OPTION before MOVE = max speed + max accel; OPTION during move = speed boost only.

Soft limits (A / B window)

ACTION	KEY	DESCRIPTION
SET + MOVE_L tap	S < tap	Set left end here on selected axes
SET + MOVE_R tap	S > tap	Set right end here on selected axes
SET + MOVE_L hold ≥ 1s	S < hold ≥ 1s	Reset L → envelope min (or –2000)
SET + MOVE_R hold ≥ 1s	S > hold ≥ 1s	Reset R → envelope max (or +2000)
SET + L + R hold ≥ 1s	S < > hold ≥ 1s	Reset both ends on selection

Travel only inside the working window. Missing envelope side → ±2000 fallback.

SET / OPTION

ACTION	KEY	DESCRIPTION
SET tap (idle)	S tap	Nothing
SET hold ≥ 1s (idle)	S hold ≥ 1s	Accel preset L — only if accel is set
SET hold ≥ 3s (idle)	S hold ≥ 3s	Accel preset H — only if set
SET hold ≥ 5s (idle)	S hold ≥ 5s	Accel learn from SPEED pot — only if set
OPTION + SET tap	* S tap	Disable driver
OPTION + SET hold ≥ 1s	* S hold ≥ 1s	Toggle loop armed
Any key while disabled	–	Re-enable driver

Loop arm does not start motion; next MOVE cruise ping-pongs until SET / same tip / halt.

Speed / accel

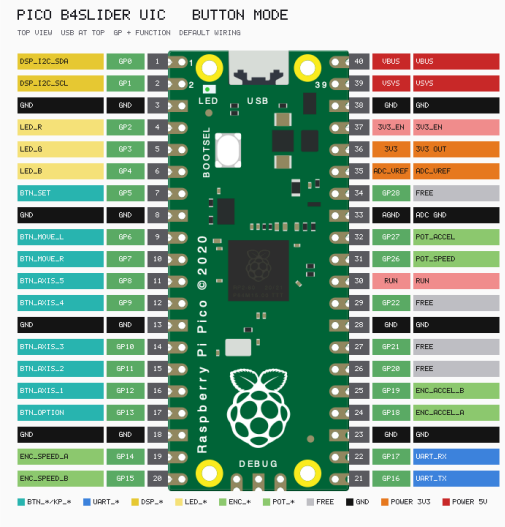
CONTROL	DESCRIPTION
SPEED pot / rotary	B4S_SPEED_INPUT pot (default) or rotary. Rotary boot vmax/8; clamp [vmax/128, vmax]; green blink on clamp enter.
ACCEL pot / rotary / set	B4S_ACCEL_INPUT. Pico ENC 14/15 + 18/19; Zero ENC 24/25 + 22/23.

Workflows

- Normal moving
 - Unlock with OPTION if LED is rainbow
 - Select AXIS (or leave 1); dial SPEED
 - Tap MOVE to cruise; hold MOVE to jog
 - Same tip stops; opposite reverses; SET stops; L+R halt
- Frame A / B on selection
 - Hold the AXIS keys you are framing
 - Jog to first pose; S < tap → left ends
 - Jog to second pose; S > tap → right ends
 - Rehearse: MOVE tap cruises to that end
 - Optional loop: idle * S hold ≥ 1s, then MOVE tap

Status LED

COLOUR	MEANING
Rainbow	Power-on / unlock
Dim white	Idle, ready
White ↔ blue	Loop armed (idle)
Yellow	Accel / decel
Green	Steady speed
+30% / 100% blue	Near / at soft end
+10% blue	Loop running
Dim orange	Driver off
Red fast blink	Hard limit
Red blink	Homing
Red	DRV_ERROR / E-stop
White flash ×N	AXIS selection changed (N ≤ 5)
Blue blink	Illegal AXIS chord
Green blink	Encoder clamp enter
White blip	Soft limit set / reset
White flash /s	SET hold second counter
Violet ×1 / ×2 / ×3	Accel preset L / H / learn
Red flash	Halt / all-four reset



UART to SliderMC @ 115200 baud (GP16/17 crossed). AXIS 12/11/10/9/8 · ENC 14/15 and 18/19 · GP22 free. Handshake: UIC \n → MC welcome # ... banner.